UAPI testing with KUnit

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Background: Testing the generic vDSO

- Started working on the vDSO
- Migrating architecture-specific code to a generic library
- Needs to be tested, preferably with the existing selftests
- No special support for cross-compilation in kselftests
- Experimentation for a usable test setup
  - nolibc-test provides cross-architecture scaffolding
  - Manually transplant vDSO selftests
  - Ugly, not upstreamable

Kunit UAPI tests

- Integration of kselftests into KUnit
- Build kselftests as `userprogs`
- Embed the test binaries in the kernel image or module
- Use KUnit to drive the test execution
- Preinit executable which provide a standard environment
- Embed the TAP output from kselftests into KUnit's KTAP
- Unified execution and reporting

nolibc

- Regular userprogs require a userspace toolchain
- Often not available when building a kernel
- nolibc can fill in the missing libc parts
- A predictable base, always in sync with the kernel
- For the vDSO kselftests no invasive changes were necessary

Required implementation

- ~400 LOC for the KUnit UAPI framework
- kbuild wrappers around the kselftests
- KUnit object embedding the kselftests and defining KUnit structure
- New kconfig symbols
- `.kunitconfig`

vDSO selftest execution

```
$ kunit.py run --kunitconfig lib/vdso/tests --arch x86_64 vdso
Running tests with:
$ qemu-system-x86_64 (...)
==================== vdso (6 subtests) ====================
====================== (29 subtests) ======================
[PASSED] __vdso_gettimeofday
[PASSED] __vdso_clock_gettime CLOCK_REALTIME
(...)
[PASSED] __vdso_time
====================== [PASSED] abi ======================
==================== [PASSED] chacha =====================
[PASSED] correctness
[PASSED] getcpu
=================== [PASSED] getrandom ====================
[PASSED] gettimeofday
====================== [PASSED] vdso ======================
Testing complete. Ran 35 tests: passed: 26, skipped: 9
Elapsed time: 6.912s total, 0.002s configuring, 5.341s building, 1.528s running
```

Current state

- Already has helped mainlining my original vDSO changes
- Found various bugs in the vDSO selftests on some architectures
- Used by me for ongoing vDSO development
- Prototype for a combination with runtime verification (RV)

Discussion topics

- What is preventing the merge to mainline?
- Which features are missing?
- How exactly to coexist with tools/testing/selftests/?
- How to handle libgcc dependencies?
- How to handle different UAPI ABIs per architecture?
- How to handle complex test programs?
- Maybe not really unit tests anymore?
- LKML feedback: Does it bloat the kernel?

Links

- KUnit UAPI tests on LWN: https://lwn.net/Articles/1029077/
- KUnit UAPI tests on LKML: 20250717-kunit-kselftests-v5-0-442b711cde2e@linutronix.de

Thank you!

- Thomas Weißschuh <thomas.weissschuh@linutronix.de>

vDSO tests (1)

```
# lib/vdso/tests/Kconfig

config VDSO_UAPI_TEST
        tristate "vDSO UAPI tests" if !KUNIT_ALL_TESTS
        depends on KUNIT_UAPI
        default KUNIT_ALL_TESTS
```

vDSO tests (2)

```
/* lib/vdso/tests/vdso_test_abi.c */

#include "../../../tools/testing/selftests/vDSO/parse_vdso.c"
#include "../../../tools/testing/selftests/vDSO/vdso_test_abi.c"
```

vDSO tests (3)

```c
/* lib/vdso/tests/vdso-tests-main.c */
#include <kunit/test.h>
#include <kunit/uapi.h>

static void abi(struct kunit *test)
{
    KUNIT_UAPI_EMBED_BLOB(vdso_test_abi, "vdso_test_abi");
    kunit_uapi_run_kselftest(test, &vdso_test_abi);
}

static struct kunit_case vdso_test_cases[] = {
    KUNIT_CASE(abi),
    /* ... */
    {}
};

static struct kunit_suite vdso_test_suite = {
```

vDSO tests (4)

```
# lib/vdso/tests/Makefile
include $(srctree)/init/Makefile.nolibc

ccflags-y                       := -I$(obj)

obj-$(CONFIG_VDSO_UAPI_TEST)    += vdso-tests.o
vdso-tests-y                    += vdso-tests-main.o

userprogs                       += vdso_test_abi
vdso_test_abi-userccflags       := $(NOLIBC_USERCFLAGS)
vdso_test_abi-userldlibs        := $(NOLIBC_USERLDLIBS)

$(obj)/vdso-tests-main.o: $(obj)/vdso_test_abi
```

vDSO tests (5)

```
KTAP version 1
1..1
    KTAP version 1
    # Subtest: vdso
    # module: vdso_tests
    1..1
        TAP version 13
        1..29
        # [vDSO kselftest] VDSO_VERSION: LINUX_2.6
        # The time is 1765430445.207582
        ok 1 __vdso_gettimeofday
(...)
        # 9 skipped test(s) detected.
        # Totals: pass:20 fail:0 xfail:0 xpass:0 skip:9 error:0
    ok 1 abi
ok 1 vdso
```